SIMATS ENGINEERING
ASSESSMENT TOOL 3

NAME: Siva Dakshitha K **REG.NO:** 192425395
COURSE NAME: CSA1603 **COURSE CODE:** Data Warehousing and Data Mining

---

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 20%

QUESTIONS: 05 **Total Marks:50**

Annexure A

COMPARITIVE ANALYSIS

Criteria	Scope of Items *(2)*	Comparison Depth(2.5)	Use of Evidence *(2)*	Critical Thinking *(2)*	Clarity of Presentation *(1)*	*Total(10)*
Weightage	*20%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I _Siva Dakshitha K / 192425395_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Siva Dakshitha K
Signature of the student:

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Q1 . Dataset: Assessment Tool 3: Comparative Analysis (25% → 5 Marks)

Time duration :45 Minutes
Assessment Tool 3 (Low)

Q1 . Dataset: Customer Data

Customer	Income (₹)	Spending Score
C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

Question:
Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. *(BL3 – 1 Mark)*

Q1) K Means Vs Hierarchical clustering

⇒ K - Means clustering divides the customers into a fixed no. of clusters using centroids.

⇒ It is fast and works well for large datasets.

⇒ It repeatedly assigns customers to the nearest centroid and updates the centroid

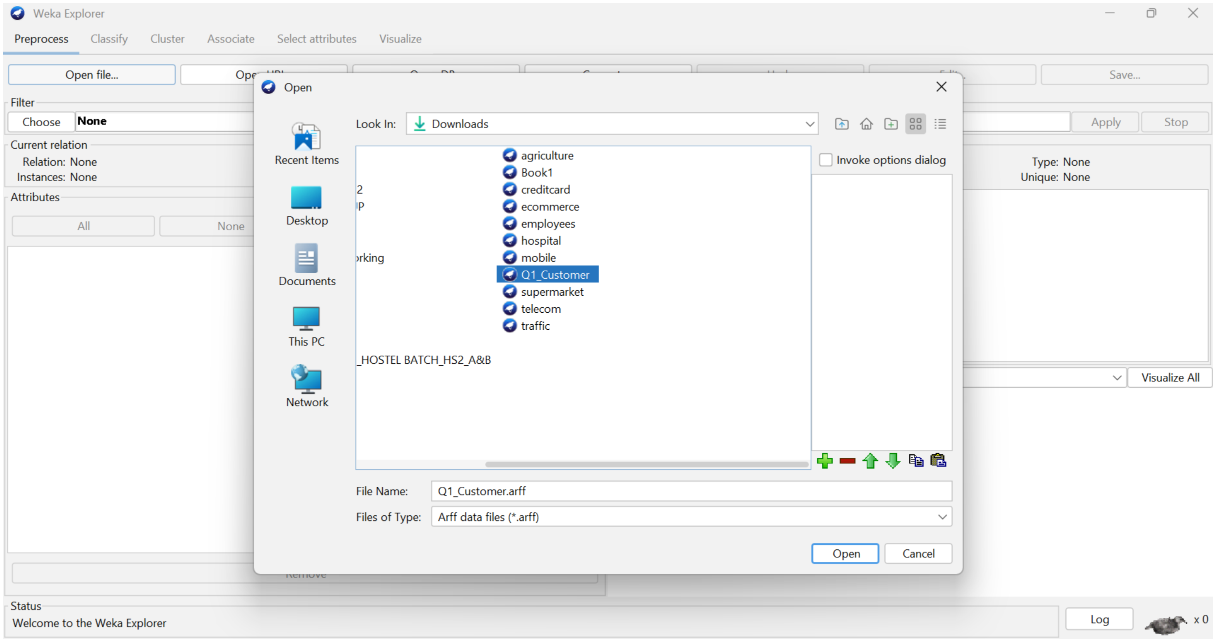
⇒ Hierarchical clustering creates clusters step-by-step.

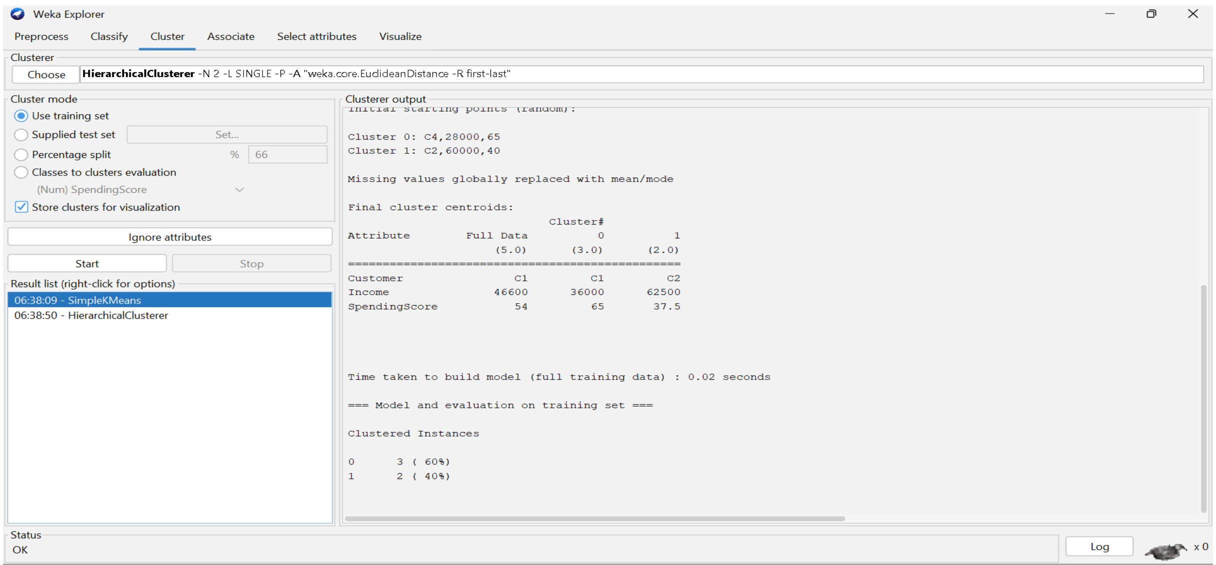
⇒ It uses a dendrogram to show relationships between customers.

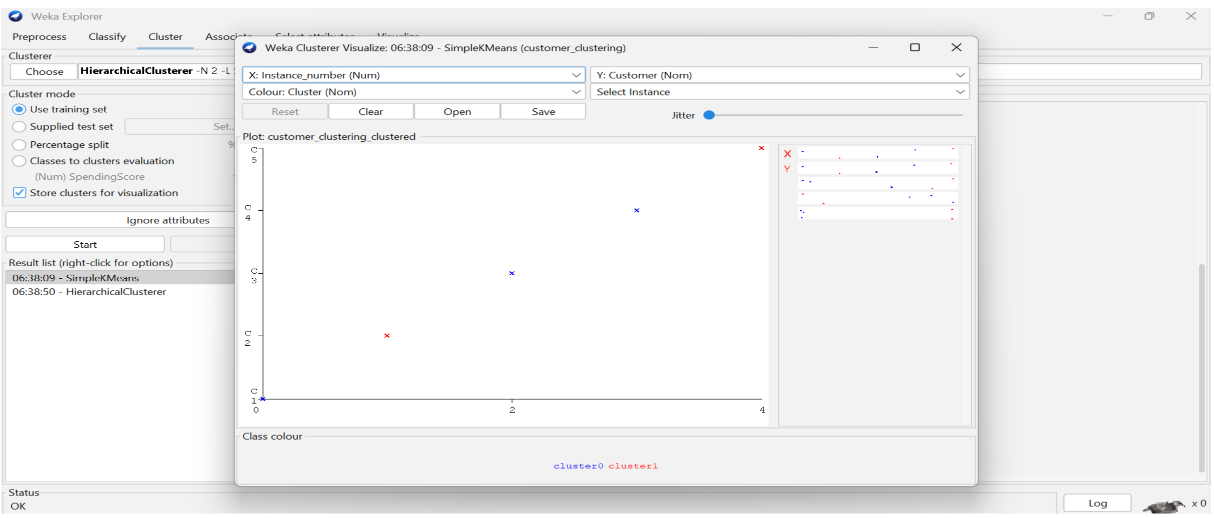
⇒ It is slower and requires more computation for large datasets.

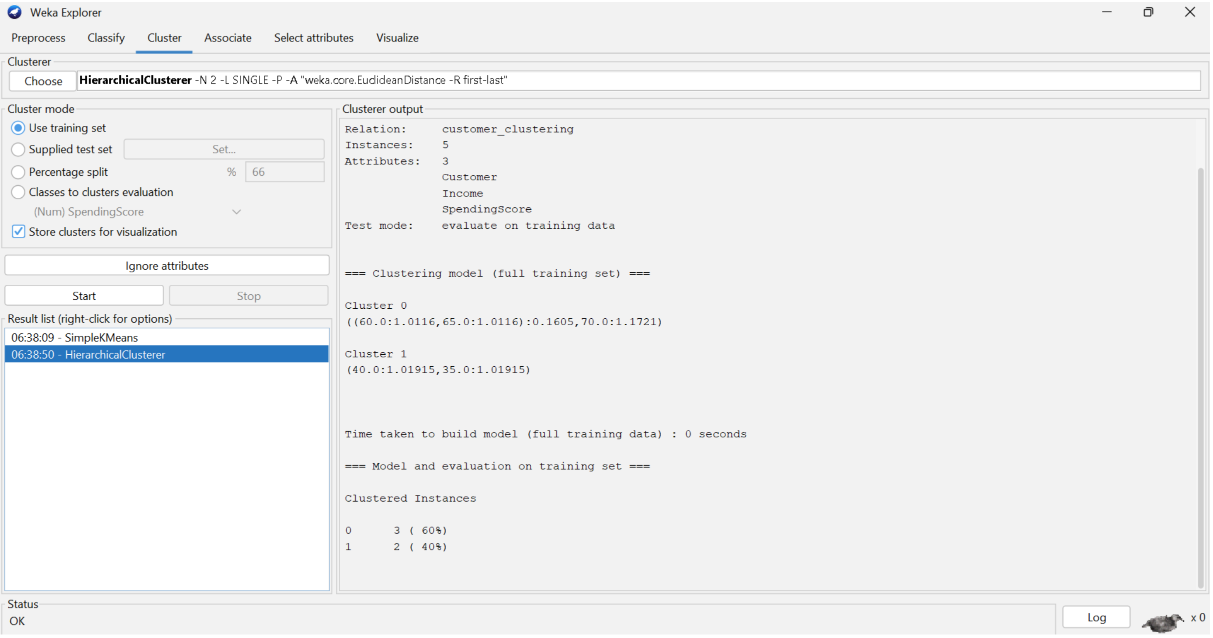
Conclusion:
K-means is better for large datasets customer data, while hierarchical clustering is useful for understanding cluster relationships.

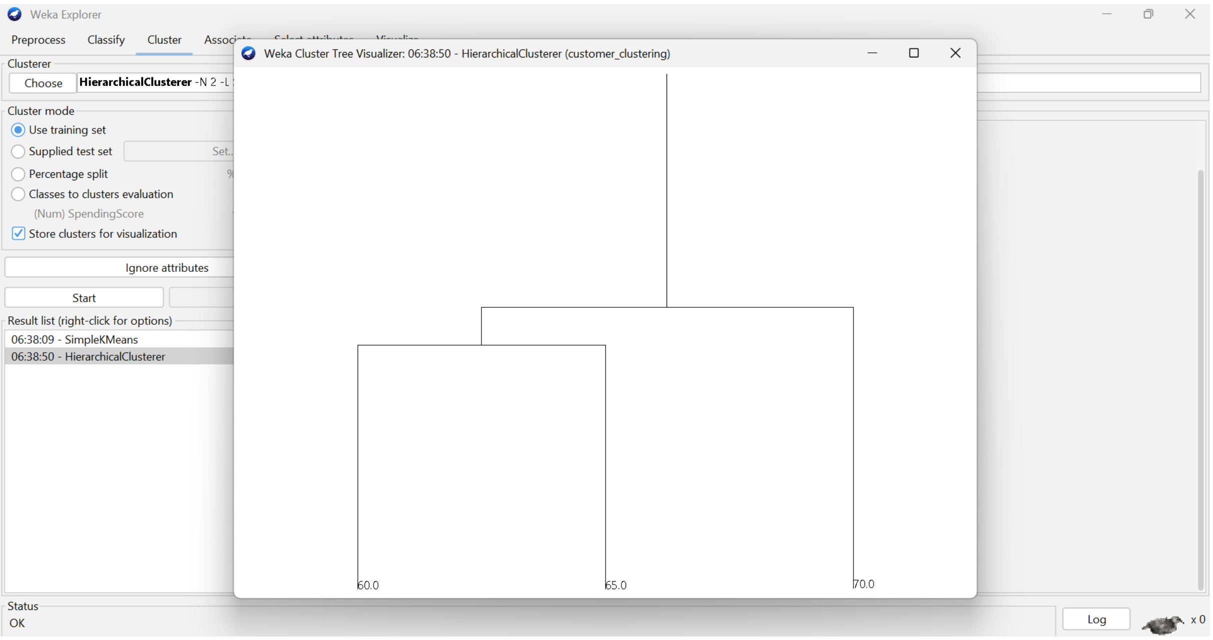
OUTPUT:











Q2. Dataset: Document Data

Document	Word1	Word2	Word3
D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question:

Compare K-means and EM algorithms in handling overlapping clusters for the given dataset. *(BL4 – 1 Mark)*

Q2)

K - Means Vs EM - Algorithms

→ K-Means assigns each document to only one cluster.

→ It works better when clusters are clearly separated.

→ EM gives a probability of a document belonging to different clusters

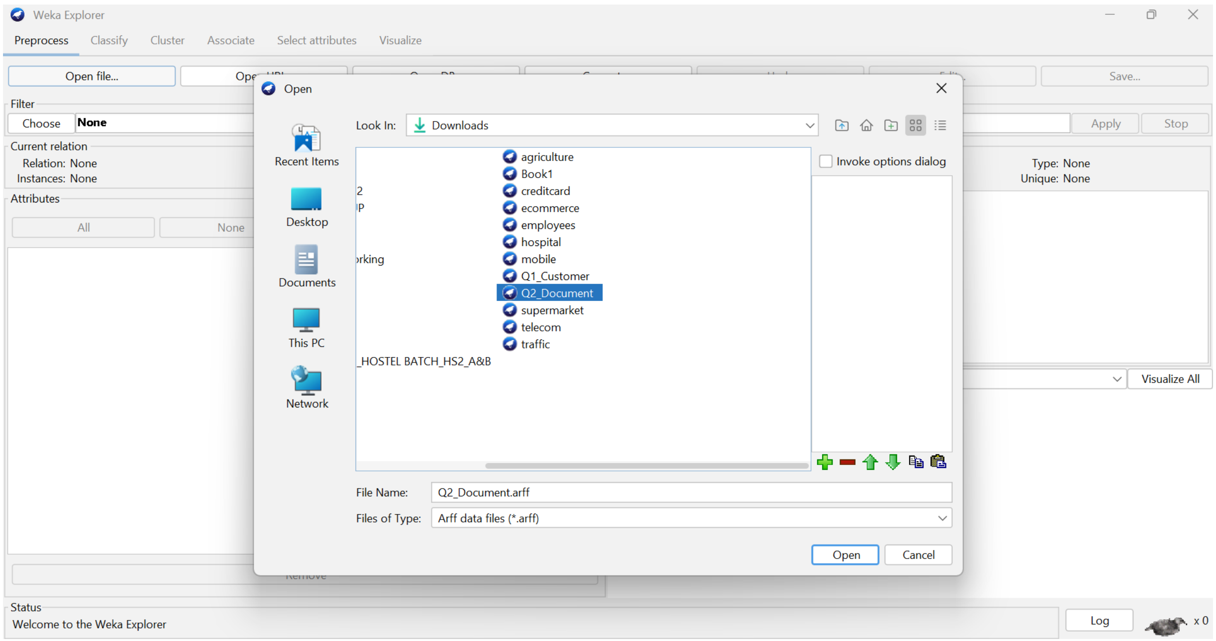
→ Therefore, a document containing words from two topics can have membership in both clusters.

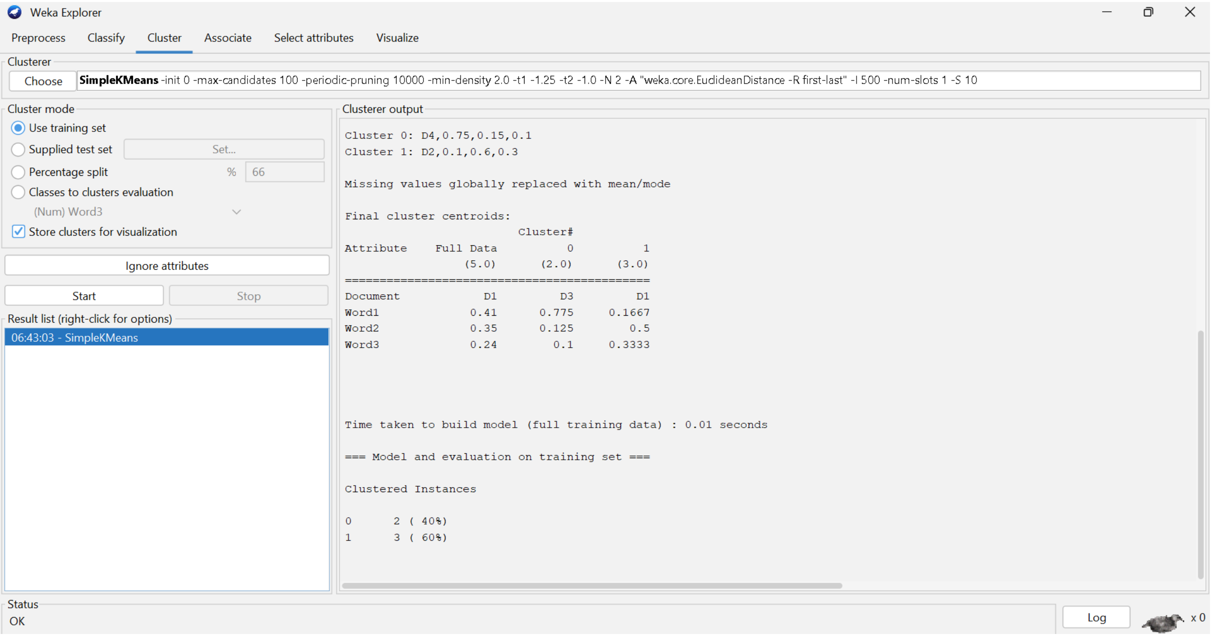
→ This is useful for the given document data because some documents may have overlapping word patterns.

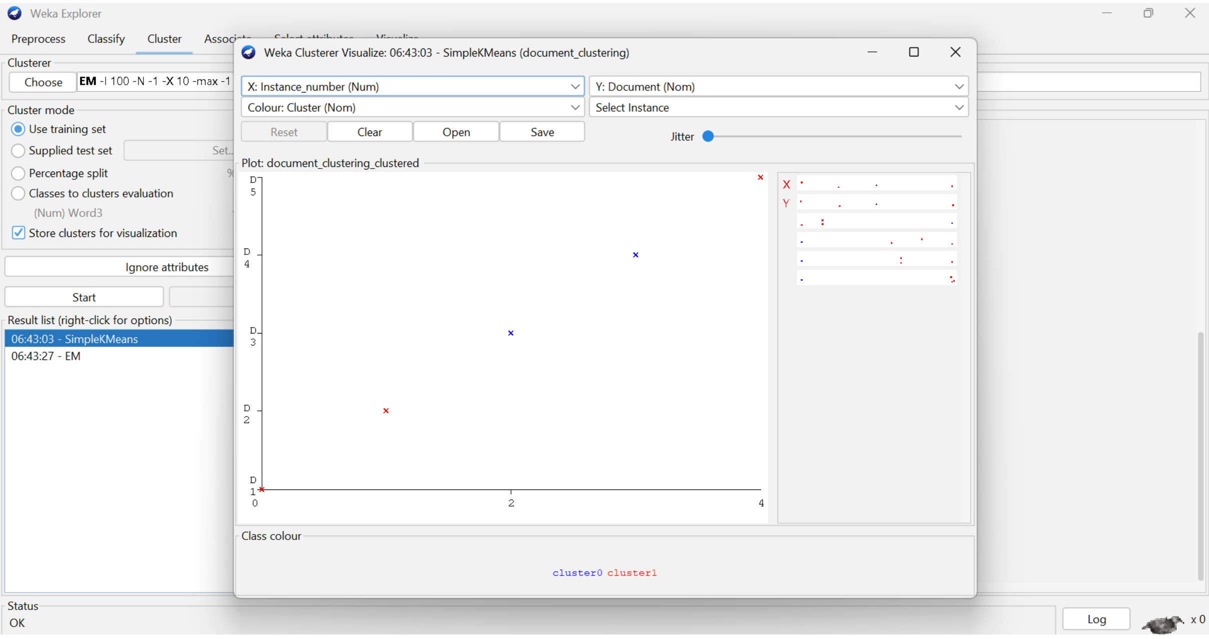
Conclusion :

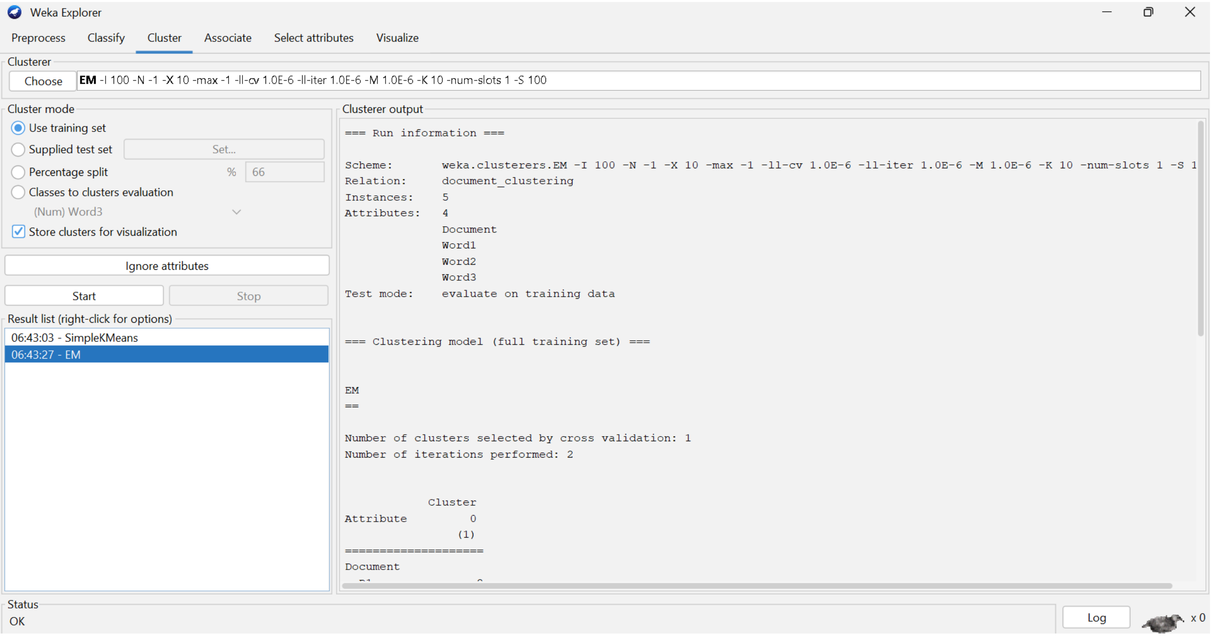
EM is better than K-Means for handling overlapping document clusters.

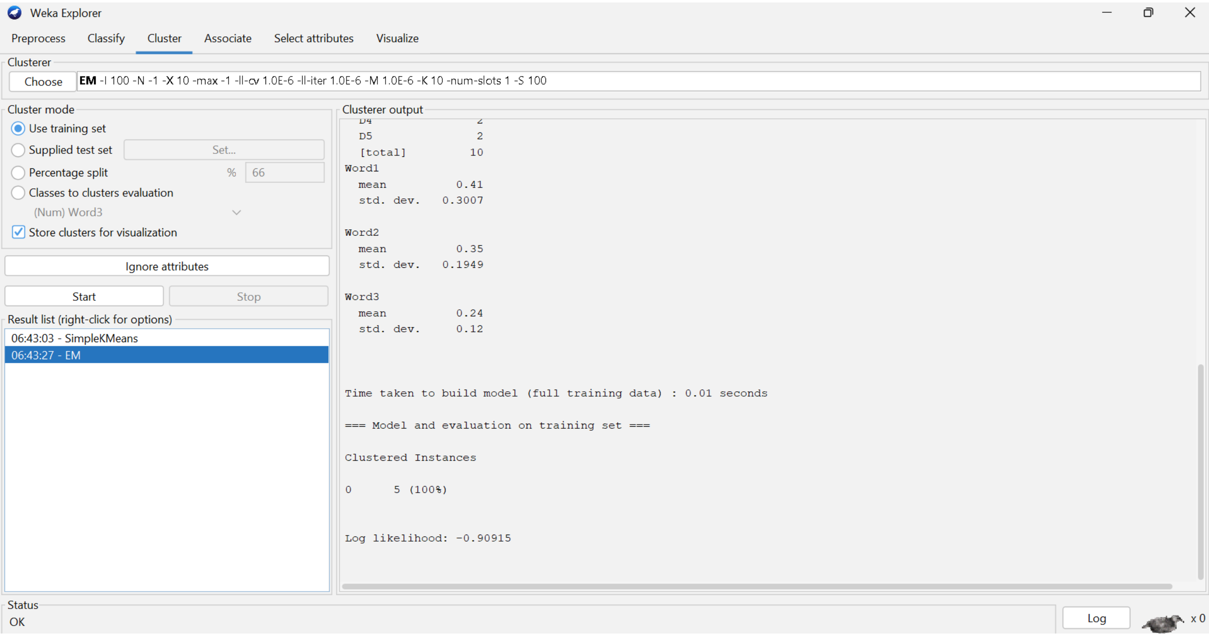
OUTPUT:

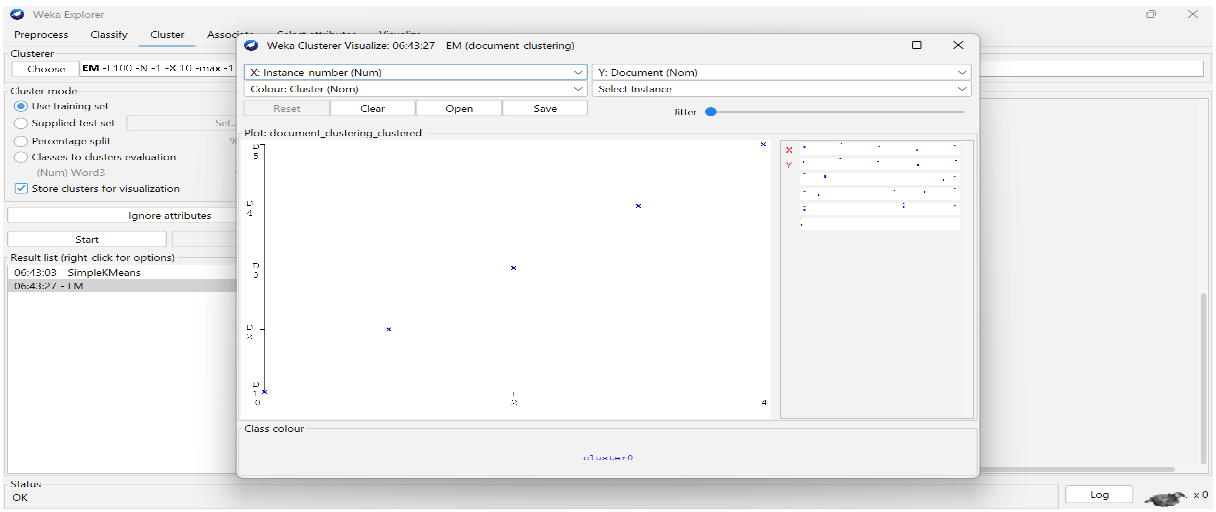



Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Word3
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:43:03 - SimpleKMeans
Clusterer output
Cluster 0: D4,0.75,0.15,0.1
Cluster 1: D2,0.1,0.6,0.3
Missing values globally replaced with mean/mode
Final cluster centroids:
Cluster#
Attribute Full Data 0 1
(5.0) (2.0) (3.0)
Document D1 D3 D1
Word1 0.41 0.775 0.1667
Word2 0.35 0.125 0.5
Word3 0.24 0.1 0.3333
Time taken to build model (full training data) : 0.01 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 2 (40%)
1 3 (60%)
Status
OK
Log
x 0


Weka Explorer
Preprocess
Classify
Cluster
Clusterer
Choose
EM -I 100 -N -1 -X 10 -max -1
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
(Num) Word3
Store clusters for visualization
Ignore attributes
Start
Result list (right-click for options)
06:43:03 - SimpleKMeans
06:43:27 - EM
Weka Clusterer Visualize: 06:43:03 - SimpleKMeans (document_clustering)
X: Instance_number (Num)
Y: Document (Nom)
Colour: Cluster (Nom)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: document_clustering_clustered
D 5
D 4
D 3
D 2
D 1
0
2
4
X
Y
Class colour
cluster0 cluster1
Status
OK
Log
x 0


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
EM -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Word3
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:43:03 - SimpleKMeans
06:43:27 - EM
Clusterer output
=== Run information ===
Scheme: weka.clusterers.EM -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 1
Relation: document_clustering
Instances: 5
Attributes: 4
Document
Word1
Word2
Word3
Test mode: evaluate on training data
=== Clustering model (full training set) ===
EM
==
Number of clusters selected by cross validation: 1
Number of iterations performed: 2
Cluster
Attribute 0
(1)
===================
Document
Status
OK
Log
x 0


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
EM -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Word3
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:43:03 - SimpleKMeans
06:43:27 - EM
Clusterer output
D5 2
[total] 10
Word1
mean 0.41
std. dev. 0.3007
Word2
mean 0.35
std. dev. 0.1949
Word3
mean 0.24
std. dev. 0.12
Time taken to build model (full training data) : 0.01 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 5 (100%)
Log likelihood: -0.90915
Status
OK
Log
x 0


Weka Explorer
Weka Clusterer Visualize: 06:43:27 - EM (document_clustering)
X: Instance_number (Num)
Y: Document (Nom)
Colour: Cluster (Nom)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: document_clustering_clustered
D5
D4
D3
D2
D1
0
2
4
X
Y
Class colour
cluster0
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
(Num) Word3
Store clusters for visualization
Ignore attributes
Start
Result list (right-click for options)
06:43:03 - SimpleKMeans
06:43:27 - EM
Status
OK
Log
x 0

Q3. Dataset: Geographic Data

Point	Latitude	Longitude
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

Question:

Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. *(BL4 – 1 Mark)*

Q3)

Geographic Data – Euclidean vs Manhattan

Given :

P1 – P5 have latitude values from 12.97 – 13.10 and longitude values from 80.20 – 80.30

- Ecludiean distance measures the direct straight line distance between two geographic points.
- Formula :

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

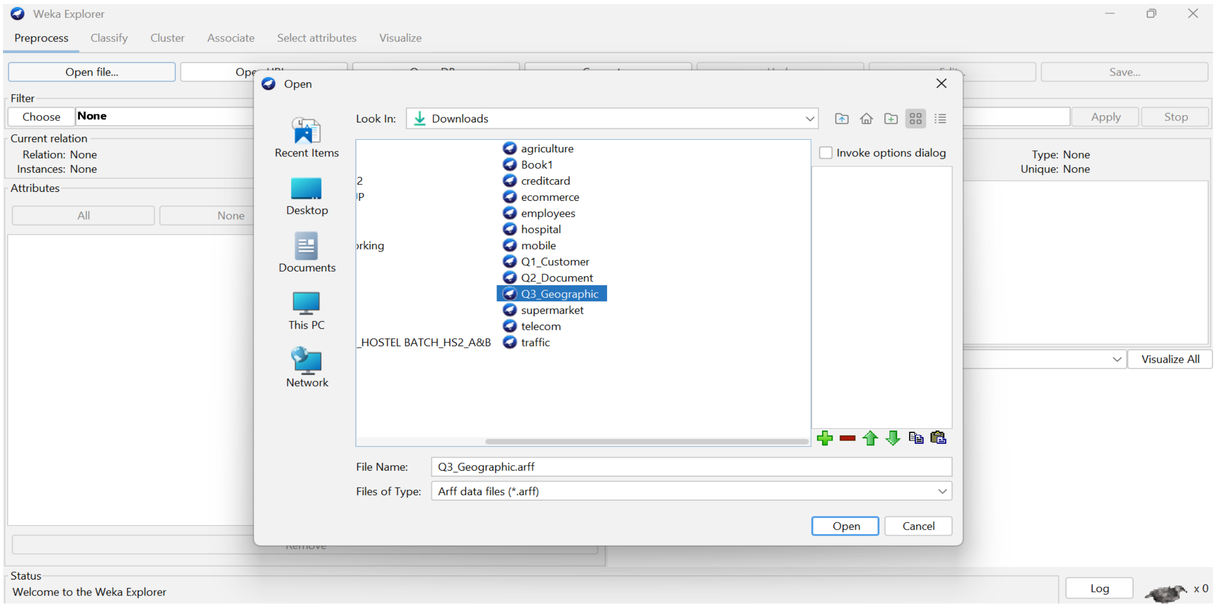
- Manhattan distance adds the absolute latitude and longitude differences
- Formula :

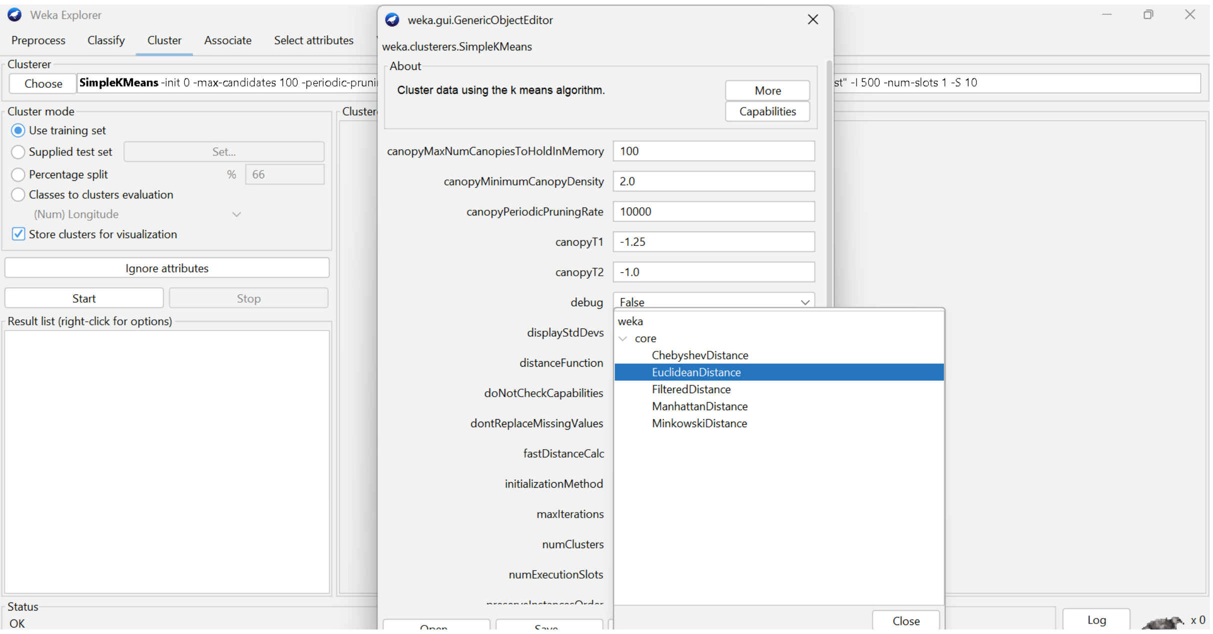
$$d = |x_1 - x_2| + (y_1 - y_2)$$

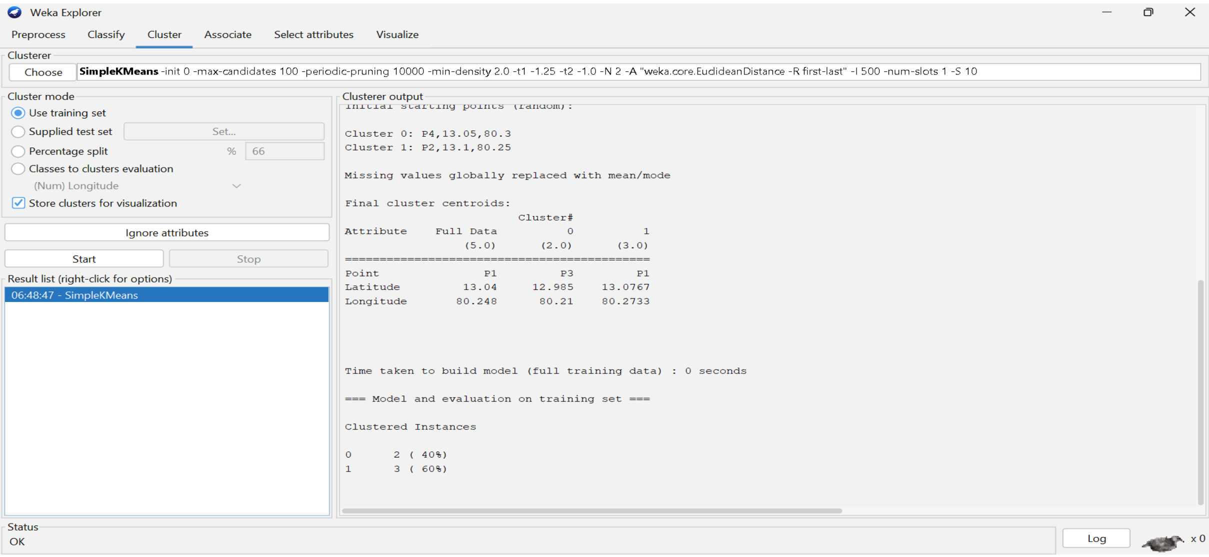
- Conclusion :

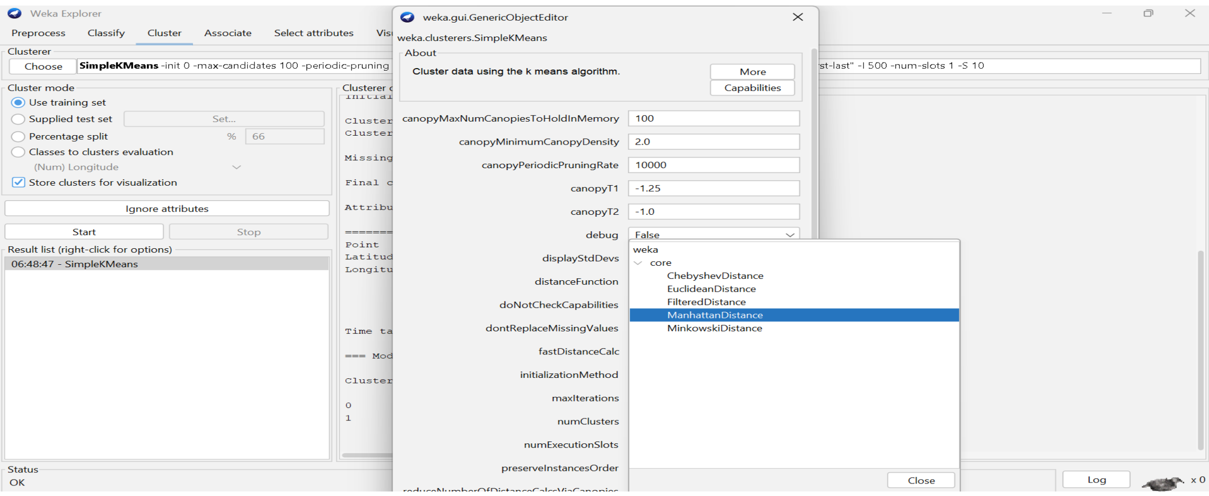
Euclidean distance is more suitable here because the data represents geographic locations and direct spatial closeness is important.

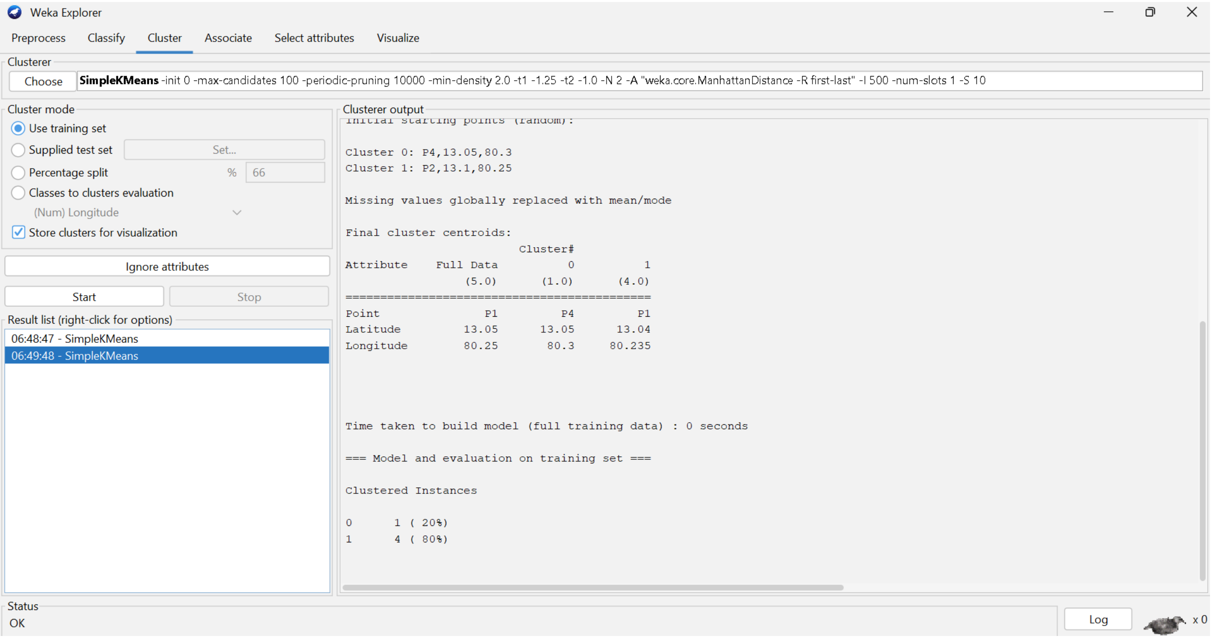
OUTPUT:

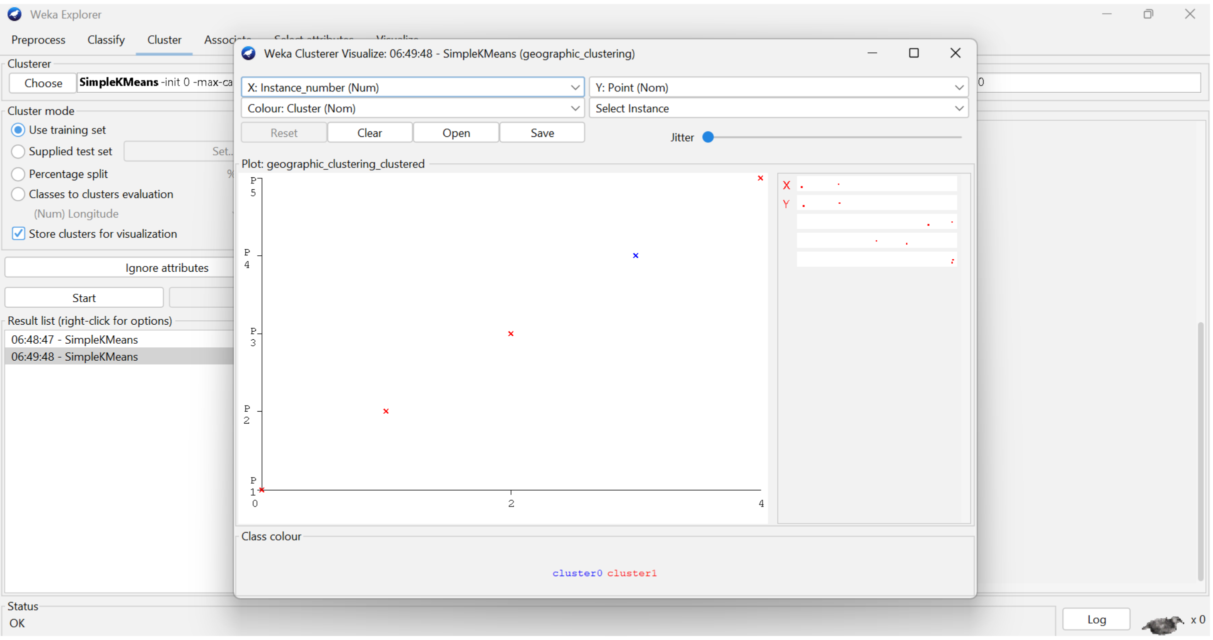



Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Clusterer
Choose
SimpleKMeans -init 0 -max-candidates 100 -periodic-pruni
st" -I 500 -num-slots 1 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Longitude
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
Status
OK
weka.gui.GenericObjectEditor
weka.clusterers.SimpleKMeans
About
Cluster data using the k means algorithm.
More
Capabilities
canopyMaxNumCanopiesToHoldInMemory 100
canopyMinimumCanopyDensity 2.0
canopyPeriodicPruningRate 10000
canopyT1 -1.25
canopyT2 -1.0
debug False
displayStdDevs
distanceFunction
doNotCheckCapabilities
dontReplaceMissingValues
fastDistanceCalc
initializationMethod
maxIterations
numClusters
numExecutionSlots
weka
core
ChebyshevDistance
EuclideanDistance
FilteredDistance
ManhattanDistance
MinkowskiDistance
Close
Log
x 0


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Longitude
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:48:47 - SimpleKMeans
Clusterer output
Initial starting points (random):
Cluster 0: P4,13.05,80.3
Cluster 1: P2,13.1,80.25
Missing values globally replaced with mean/mode
Final cluster centroids:
Cluster#
Attribute Full Data 0 1
(5.0) (2.0) (3.0)
Point P1 P3 P1
Latitude 13.04 12.985 13.0767
Longitude 80.248 80.21 80.2733
Time taken to build model (full training data) : 0 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 2 (40%)
1 3 (60%)
Status
OK
Log
x 0


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Clusterer
Choose
SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning
st-last" -I 500 -num-slots 1 -S 10
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Longitude
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:48:47 - SimpleKMeans
Status
OK
weka.gui.GenericObjectEditor
weka.clusterers.SimpleKMeans
About
Cluster data using the k means algorithm.
More
Capabilities
canopyMaxNumCanopiesToHoldInMemory 100
canopyMinimumCanopyDensity 2.0
canopyPeriodicPruningRate 10000
canopyT1 -1.25
canopyT2 -1.0
debug False
displayStdDevs
distanceFunction
doNotCheckCapabilities
dontReplaceMissingValues
fastDistanceCalc
initializationMethod
maxIterations
numClusters
numExecutionSlots
preserveInstancesOrder
weka
core
ChebyshevDistance
EuclideanDistance
FilteredDistance
ManhattanDistance
MinkowskiDistance
Close
Log
x 0





Q4. Dataset: Retail Products

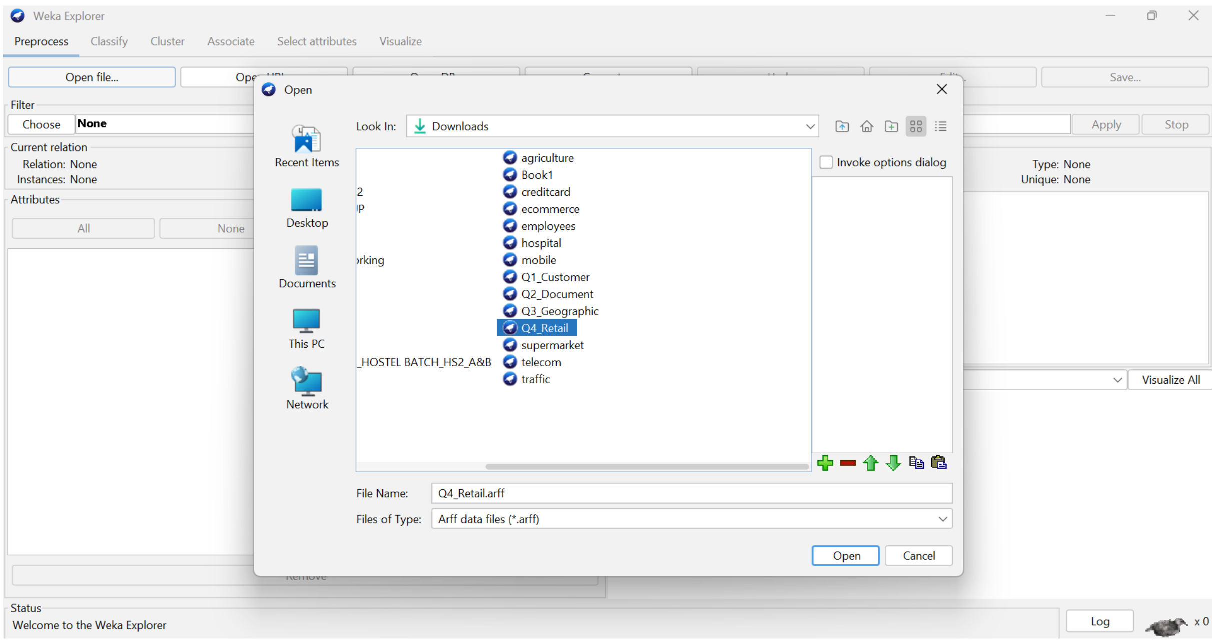
Product	Sales (₹)	Frequency
P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

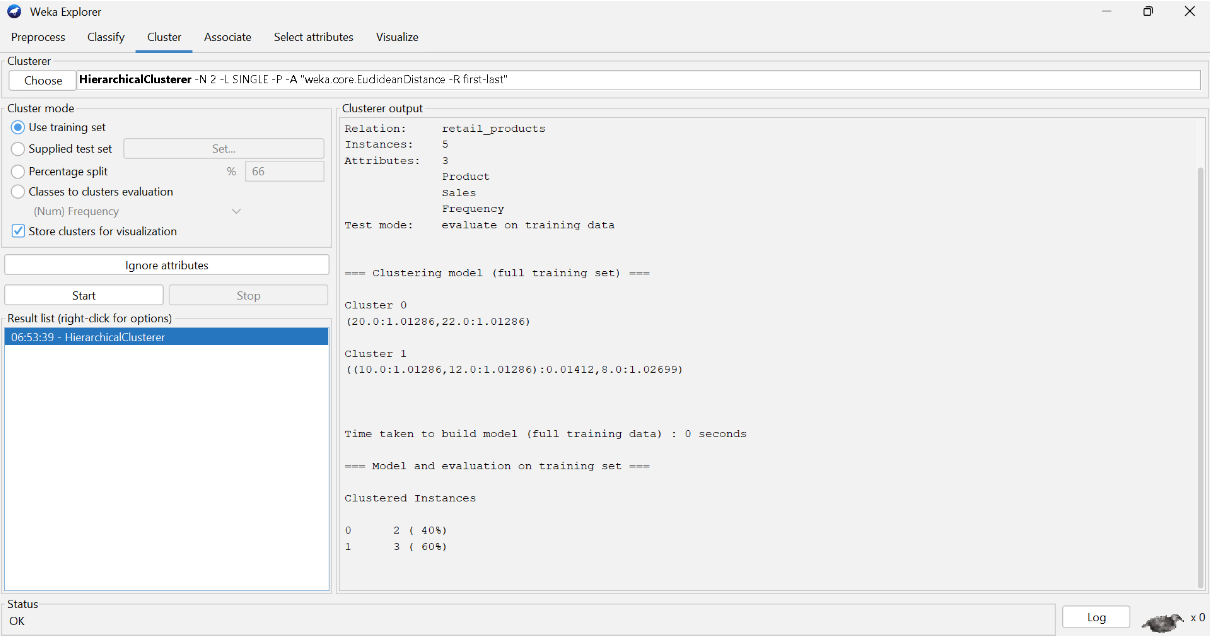
Question:
Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. *(BL3 – 1 Mark)*

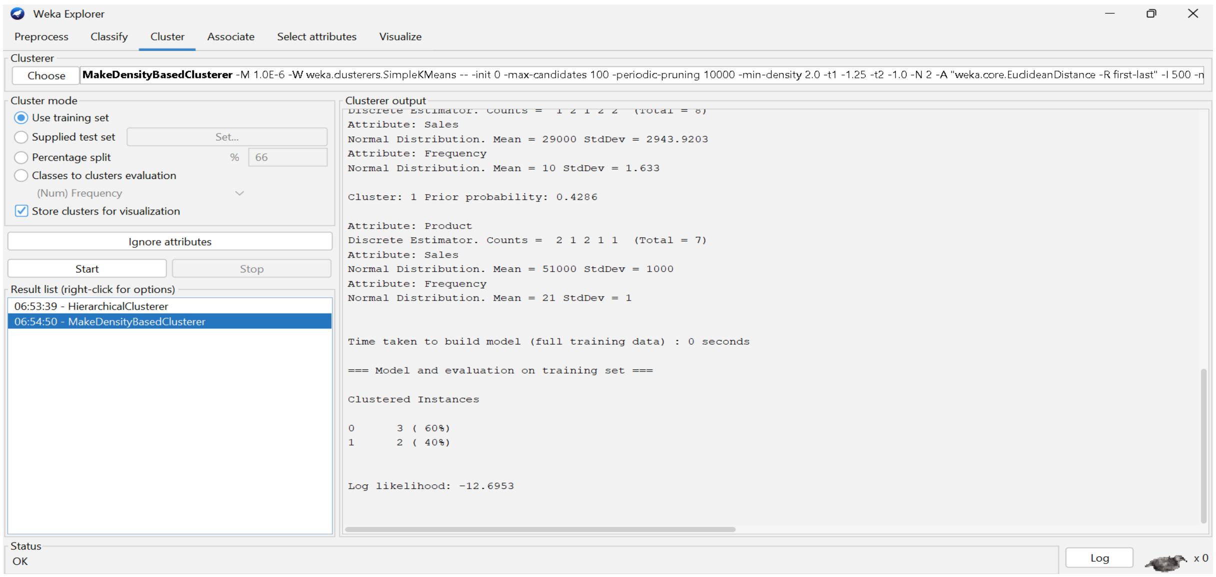
Q4)
Retail Products - Agglomerative VS Divisive
Given Sales ₹20,000 - ₹52,000 and frequency 5 to 22.

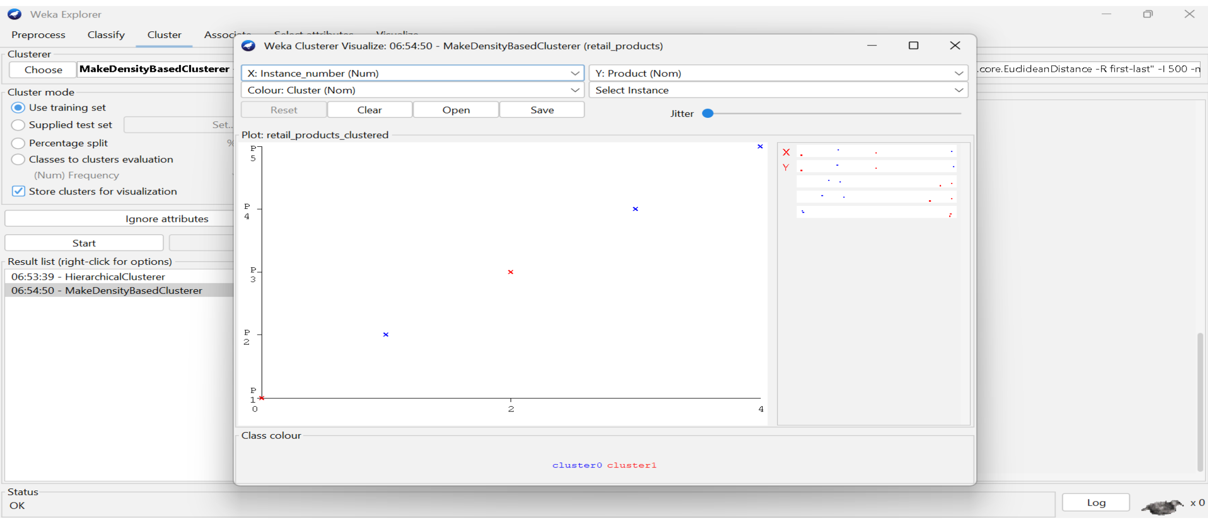
- Agglomerative clustering starts with each product as a separate cluster.
- It gradually merges similar products based on sales and frequency.
- For example, P2 (₹50,000, 22) and P4 (₹52,000, 12) have high sales and can be considered relatively similar.
- Divisive clustering starts with all five products in one cluster.
- It then splits the products into smaller groups.
- Agglomerative clustering is simple and suitable for grouping these retail products based on sales and purchase frequency.

OUTPUT:




Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize
Clusterer
Choose HierarchicalClusterer -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Frequency
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:53:39 - HierarchicalClusterer
Clusterer output
Relation: retail_products
Instances: 5
Attributes: 3
Product
Sales
Frequency
Test mode: evaluate on training data
=== Clustering model (full training set) ===
Cluster 0
(20.0:1.01286,22.0:1.01286)
Cluster 1
((10.0:1.01286,12.0:1.01286):0.01412,8.0:1.02699)
Time taken to build model (full training data) : 0 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 2 (40%)
1 3 (60%)
Status
OK
Log
x 0


Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize
Clusterer
Choose MakeDensityBasedClusterer -M 1.0E-6 -W weka.clusterers.SimpleKMeans -- -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -n
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Frequency
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:53:39 - HierarchicalClusterer
06:54:50 - MakeDensityBasedClusterer
Clusterer output
Attribute: Sales
Normal Distribution. Mean = 29000 StdDev = 2943.9203
Attribute: Frequency
Normal Distribution. Mean = 10 StdDev = 1.633
Cluster: 1 Prior probability: 0.4286
Attribute: Product
Discrete Estimator. Counts = 2 1 2 1 1 (Total = 7)
Attribute: Sales
Normal Distribution. Mean = 51000 StdDev = 1000
Attribute: Frequency
Normal Distribution. Mean = 21 StdDev = 1
Time taken to build model (full training data) : 0 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 3 (60%)
1 2 (40%)
Log likelihood: -12.6953
Status
OK
Log
x 0


Weka Explorer
Weka Clusterer Visualize: 06:54:50 - MakeDensityBasedClusterer (retail_products)
X: Instance_number (Num)
Y: Product (Nom)
Colour: Cluster (Nom)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: retail_products_clustered
P5
P4
P3
P2
P1
0
2
4
Class colour
cluster0 cluster1
Result list (right-click for options)
06:53:39 - HierarchicalClusterer
06:54:50 - MakeDensityBasedClusterer
Status
OK
Log
x 0

Q5 . Dataset: Mixed Attributes

Item	Price (₹)	Category	Rating
I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Question:
Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. *(BL4 – 1 Mark)*

Q5)
Mixed Attributes — Centroid - Based Vs Density-Based.

Given : Price → ₹250 - ₹550,
Category = A/B,
Rating = 3.5 – 4.7

→ The dataset contains numerical attributes such as price and Rating.

→ It also contains a categorical attributes, category (A/B).

→ Centroid - based clustering, such as K-Means forms groups around a central point.

→ K-Means mainly works with numerical values, so category must be properly encoded.

→ Density - based clustering, such as DBSCAN, forms clusters based on dense groups of products.

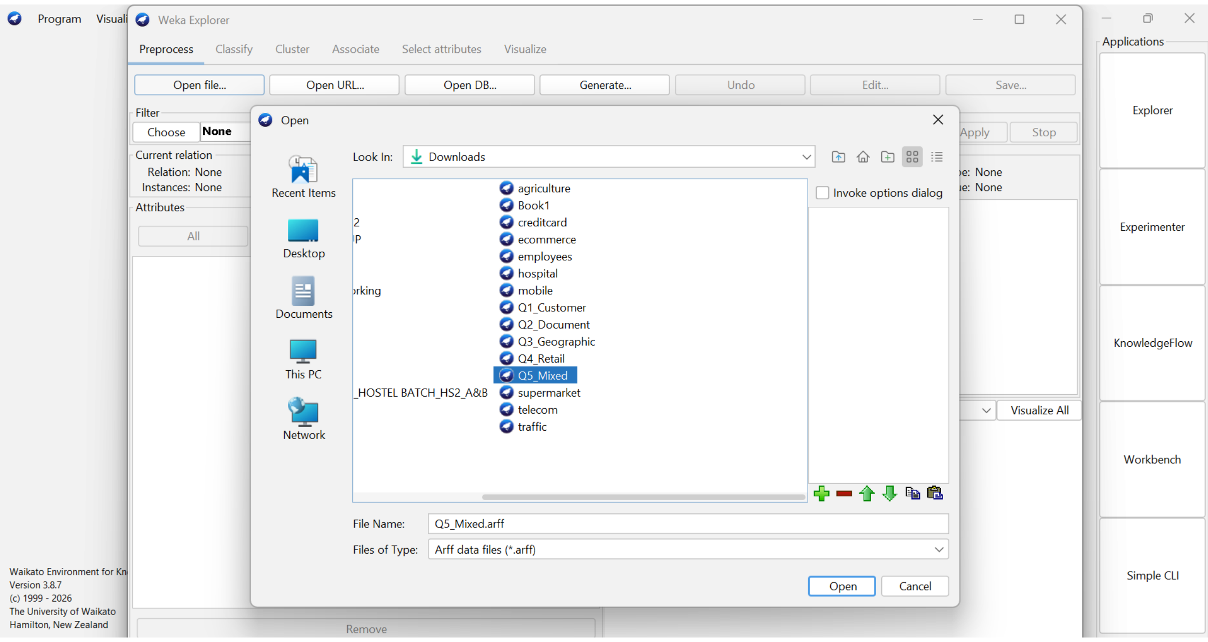
→ It can also identify unusual products as outliers.

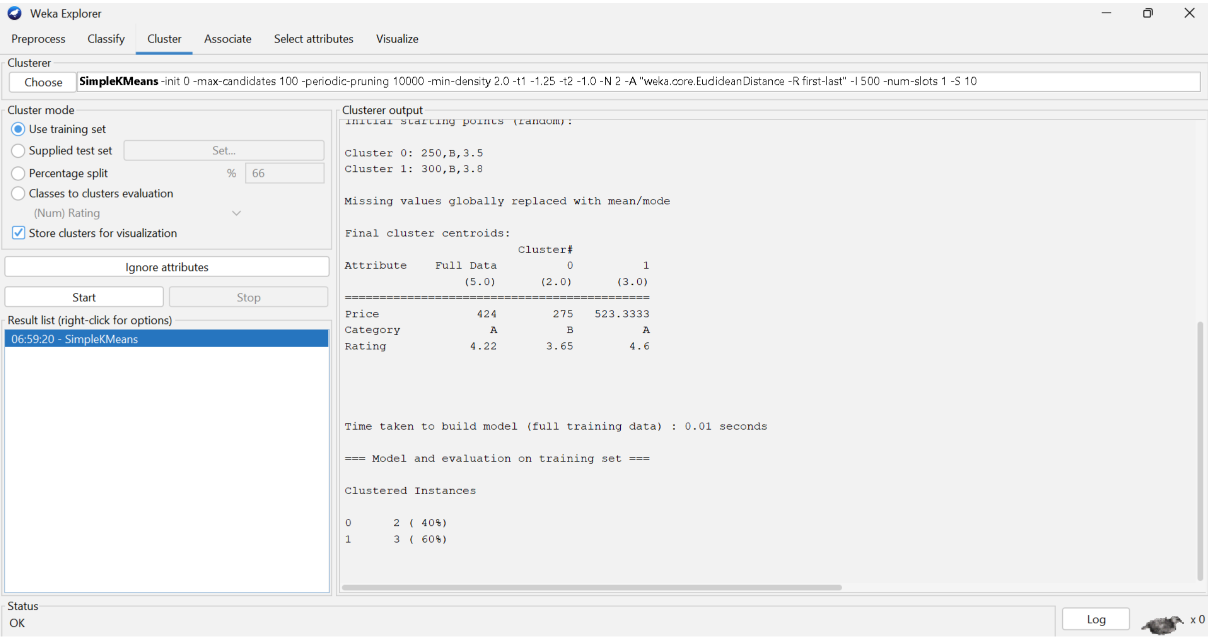
→ For this mixed dataset, suitable encoding and distance measurement are required

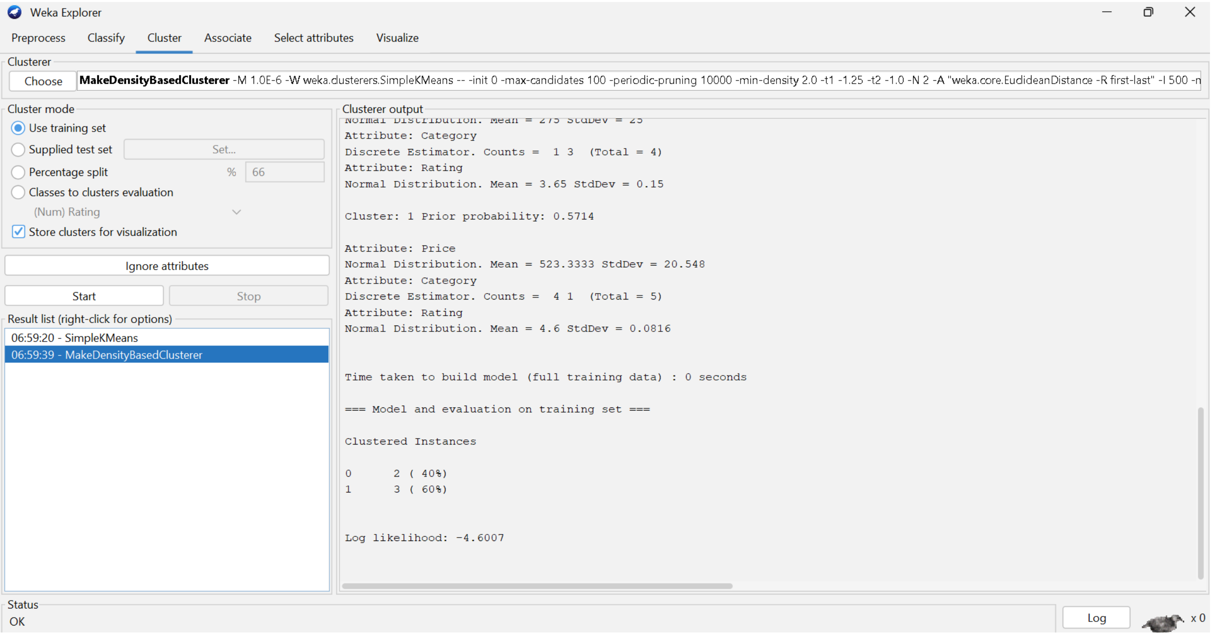
Conclusion:

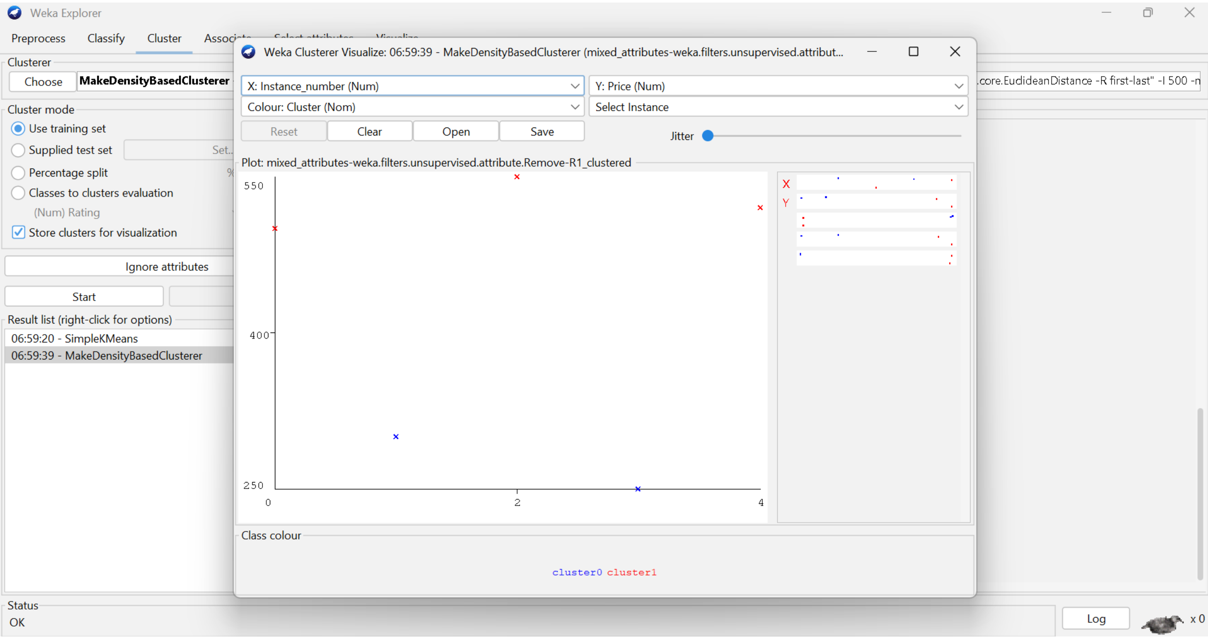
Centroid - based clustering is simple for numerical attributes while density-based clustering is useful when the products form irregular groups or contain outliers.

OUTPUT:


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Open file...
Open URL...
Open DB...
Generate...
Undo
Edit...
Save...
Open
Look In: Downloads
Recent Items
Desktop
Documents
This PC
Network
agriculture
Book1
creditcard
ecommerce
employees
hospital
mobile
Q1_Customer
Q2_Document
Q3_Geographic
Q4_Retail
Q5_Mixed
supermarket
telecom
traffic
Invoke options dialog
File Name: Q5_Mixed.arff
Files of Type: Arff data files (*.arff)
Open
Cancel
Applications
Explorer
Experimenter
KnowledgeFlow
Workbench
Simple CLI
Waikato Environment for Kn
Version 3.8.7
(c) 1999 - 2026
The University of Waikato
Hamilton, New Zealand


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
(Num) Rating
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:59:20 - SimpleKMeans
Clusterer output
Initial starting points (random):
Cluster 0: 250,B,3.5
Cluster 1: 300,B,3.8
Missing values globally replaced with mean/mode
Final cluster centroids:
Cluster#
Attribute Full Data 0 1
(5.0) (2.0) (3.0)
Price 424 275 523.3333
Category A B A
Rating 4.22 3.65 4.6
Time taken to build model (full training data) : 0.01 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 2 (40%)
1 3 (60%)
Status
OK
Log


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
MakeDensityBasedClusterer -M 1.0E-6 -W weka.clusterers.SimpleKMeans -- -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -n
Cluster mode
Use training set
Supplied test set
Set...
Percentage split
% 66
Classes to clusters evaluation
(Num) Rating
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:59:20 - SimpleKMeans
06:59:39 - MakeDensityBasedClusterer
Clusterer output
Attribute: Category
Discrete Estimator. Counts = 1 3 (Total = 4)
Attribute: Rating
Normal Distribution. Mean = 3.65 StdDev = 0.15
Cluster: 1 Prior probability: 0.5714
Attribute: Price
Normal Distribution. Mean = 523.3333 StdDev = 20.548
Attribute: Category
Discrete Estimator. Counts = 4 1 (Total = 5)
Attribute: Rating
Normal Distribution. Mean = 4.6 StdDev = 0.0816
Time taken to build model (full training data) : 0 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 2 (40%)
1 3 (60%)
Log likelihood: -4.6007
Status
OK
Log
x 0


Weka Explorer
Preprocess
Classify
Cluster
Clusterer
Choose
MakeDensityBasedClusterer
.core.EuclideanDistance -R first-last" -I 500 -n
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
(Num) Rating
Store clusters for visualization
Ignore attributes
Start
Result list (right-click for options)
06:59:20 - SimpleKMeans
06:59:39 - MakeDensityBasedClusterer
Weka Clusterer Visualize: 06:59:39 - MakeDensityBasedClusterer (mixed_attributes-weka.filters.unsupervised.attribut...
X: Instance_number (Num)
Y: Price (Num)
Colour: Cluster (Nom)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: mixed_attributes-weka.filters.unsupervised.attribute.Remove-R1_clustered
550
400
250
0
2
4
X
Y
Class colour
cluster0 cluster1
Status
OK
Log
x 0